GoreeCloud Search — deterministic result-quality benchmark

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Scope: Development source check only. This benchmark neither runs providers nor qualifies a release. It complements—not replaces—the seven-provider category-adapter tests and 24-adapter provider-qualification report gate.

Why this gate exists

A visually full search results page can still be poor: dozens of copies of the same asset, one provider silently failing, or many top results missing the known-relevant content. The first owned fusion primitive already deduplicates URLs and image assets and ranks source votes. This CLI gives that behavior a repeatable, privacy-preserving quality gate using **pinned synthetic data**.

Run from the Search repository root:

```
python3 -m scripts.search_quality_benchmark \
  tests/goreecloud/fixtures/search_quality/images_multi_source.json
python3 -m unittest discover -s tests/goreecloud -p 'test_search_quality_benchmark.py' -v
```

The fixture contains two invented sources and six generated image cards. Two URLs refer to duplicate image assets even though their hosting pages differ. Expected outcome: **four distinct images, two duplicate cards collapsed, two contributing sources, three relevant assets in top ten, 100% recall of three prejudged relevant assets, no missing original/media identity**. The updated fixture also requires **two sources with distinct top-ten discoveries**: merely mirroring another provider's cards no longer passes the diversity gate.

This is an offline contract assertion. The invented sources are **not** Google, Bing, Brave or any other live provider. These synthetic metrics are **not** a measurement of actual web, shopping or image quality.

Evidence contract and pass/fail semantics

Each fixture is bounded and has exactly five root fields:

Field	Meaning
development_only	Must be true; never infer live-provider or public release approval.
category	One implemented fusion category: web, images, videos, news.

Field	Meaning
	Shopping, Maps and More are roadmap only.
sources	Unique sanitized source IDs with explicit ok, empty, timeout, rate_limited, access_denied or error states. Only ok can contain candidate cards; errors may not disguise partial results as success.
relevant	A small, hand-judged set of result URLs; for Images, the asset URLs rather than hosting-page URLs.
gates	Minimum diversity, unique result count, relevant@10, recall@10, maximum images without original-media IDs, optional distinct top-ten discovery-source count (min_sources_with_unique_at_10), and optional full-source-health requirement.

The evaluator emits only aggregate counts, source IDs and fixed failure codes:

- distinct_results and collapsed_duplicates show how much the fusion engine truly adds rather than duplicating cards. contributing_sources counts distinct sources with at least one vote, not simply configured names.
- unique_results_at_10 and sources_with_unique_at_10 report how many top-ten results have only one contributing source and how many separate sources contribute at least one such distinct discovery. Two providers reporting the exact same images are *not* credited as two independent discoverers. The optional threshold is evaluated alongside existing gates.
- failed_sources and empty_sources distinguish upstream unavailability from a valid zero-result response. Missing source diversity fails a configured gate even if one source has enough cards to fill the page.
- missing_image_media_identity catches image cards that cannot be reliably deduplicated because no media URL was supplied. The example allows zero.
- relevant_at_10, recall_percent_at_10, precision_percent_at_10, and reciprocal_rank_at_10 check the ranking of a fixed relevance corpus. A recalled result elsewhere beyond top ten does not satisfy the top-ten gate.

For Videos, judged result matching now uses the same conservative canonical identity as the owned fusion engine: proven-identical YouTube watch, Shorts, live, embed and youtu.be links to an 11-character video ID count as one asset. This never follows those

links, broadens provider permissions, or assumes two different video URLs are equivalent without a validated identity.

An executable gate requires positive minimum relevance, diversity, results and recall thresholds; zero-threshold "success" cannot bypass the quality checks. No user query, URL, title, page, media data, file path or provider response body is included in the CLI output. Unexpected fixture keys (including raw queries), malformed source identities, unsafe URLs, contradictory source statuses, duplicate judgments and oversized fixtures are rejected.

CLI exit codes: 0 = all offline fixture gates passed; 1 = legitimate but poor quality; 2 = malformed/untrusted fixture; 64 = no fixture supplied. The JSON output always includes release_qualified: false.

Real result coverage still unqualified

As of this Development milestone, users have reported missing Image results and poor relevance. The existing private Search runtime can show Bing Images while other image providers may be blocked, broken, rate-limited or disabled by the selected preset. These are separate cases with different remedies, not a justification to claim more results than actually arrived.

Before the real user-visible image experience can pass acceptance:

1. Establish provider-by-provider approved use and correct egress, image use, cache, redistribution and attribution rights. Google and Bing API product changes and Brave terms require up-to-date direct verification.
2. Obtain a repeatable, explicitly approved and appropriately rate-limited provider health sample by category. Never evade denials, challenge pages, throttling or license limits to make a coverage number pass.
3. In an isolated Development implementation, record **sanitized counts** of selected, contacted, successful, empty, denied, timed out and invalid-card providers; keep all user queries and image links out of telemetry.
4. Build a lawfully obtained, privacy-reviewed and pinned relevance corpus with human judgments. Re-run this exact algorithm against admitted provider fixtures with evaluated diversity, asset dedup, recall and ranking evidence.
5. Verify the real Web/Images/Videos/News UI renders truly fused cards, distinct unavailable-source state, keyboard/touch accessibility, responsive loading and image fallback, using exact candidate screenshots and browser tests. Shopping and Maps require separate implemented contracts.
6. Finish privacy, abuse/rate controls, security, native OS packaging, release provenance, rollbacks and actual ingress/user acceptance; do not infer public readiness or Anchor status from offline quality checks.

No public traffic, NetBird policy, deployed image, API credentials, external provider request or persistent user data is touched by this benchmark.